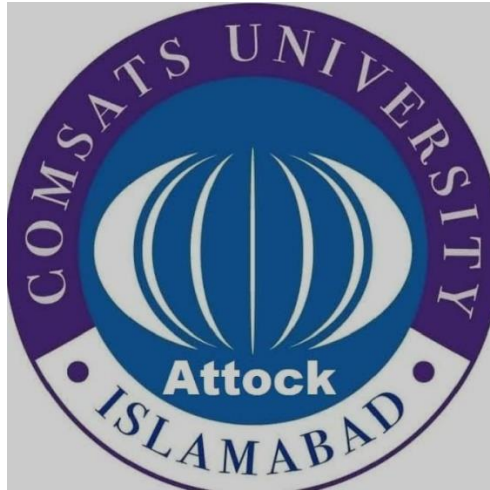




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SUBJECT	:	Information Security
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Question 1 Answer:

```
message = input("Enter message: ")
key = input("Enter single-character key: ")

ciphertext = ""
for char in message:
    ciphertext += chr(ord(char) ^ ord(key))

print("Ciphertext:", ciphertext)

decrypted = ""
for char in ciphertext:
    decrypted += chr(ord(char) ^ ord(key))

print("Decrypted text:", decrypted)
```

MidTerm-Question1 ×

⋮

C:\Users\Haider\IdeaProjects\Assignment01_Info

Enter message: *HELLO*

Enter single-character key: *A*

⏏

Decrypted text: HELLO

Process finished with exit code 0

Question 2 Answer

```
ciphertext = input("Enter ciphertext: ")
shift = int(input("Enter shift: "))

plaintext = ""

for char in ciphertext:
    if char.isalpha():
        base = ord('A') if char.isupper() else ord('a')
        plaintext += chr((ord(char) - base - shift) % 26 + base)
    else:
        plaintext += char

print("Plaintext:", plaintext)
```

MidTerm-Question2 ×

```
C:\Users\Haider\IdeaProjects\Assignment01_InformationSecurity\.venv
Enter ciphertext: khoor
Enter shift: 3
Plaintext: hello

Process finished with exit code 0
```

Question No 3

```
ciphertext = input("Enter ciphertext: ").upper()
key = input("Enter key: ").upper()

plaintext = ""
key_index = 0

for char in ciphertext:
    if char.isalpha():
        shift = ord(key[key_index % len(key)]) - ord('A')
        decrypted_char = chr((ord(char) - ord('A') - shift) % 26 + ord('A'))
        plaintext += decrypted_char
        key_index += 1
    else:
        plaintext += char

print("Plaintext:", plaintext)
```

 MidTerm-Question3 ×

```
⋮
C:\Users\Haider\IdeaProjects\Assignment01_InformationSecurity\.venv\Scripts\py
Enter ciphertext: LXFOPVEFRNHR
Enter key: LEMON
Plaintext: ATTACKATDAWN

Process finished with exit code 0
```

Question No 4

```
def caesar_encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            base = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - base + shift) % 26 + base)
        else:
            result += char
    return result

msg = input("Enter message: ")
s = int(input("Enter shift: "))
print("Ciphertext:", caesar_encrypt(msg, s))
```

 MidTerm-Question4 ×

⋮

```
C:\Users\Haider\IdeaProjects\Assignment01_InformationSecurity\.venv\
Enter message: HELLO
Enter shift: 3
Ciphertext: KH00R
```

Question No 5

→ Write one similarity between DES and AES.

Answer

Both DES and AES are symmetric key block ciphers.

They use the same key for both encryption and decryption.

Both operate on fixed-size data blocks to secure information.

→ What does CBC mode stand for in block ciphers.

Answer

CBC stands for Cipher Block Chaining mode.

In CBC, each plaintext block is XORed with the previous ciphertext block.

This makes encryption more secure by adding dependency between blocks.

→ Why is AES faster than DES.

Answer

AES is faster because it uses fewer, simpler rounds than DES.

Its algorithm is optimized for modern computer hardware.

AES also handles larger block sizes efficiently, improving speed.